

Nested array design of extended coprime sets for DOA estimation of non-circular signals

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Abstract—In recent years, direction of arrival estimation utilizing non-circular signals has become a focal point for scholarly research. To enhance the degrees of freedom (DOF) in receiver arrays specifically for non-circular signal DOA estimation, this study introduces a novel array configuration. This design leverages an extended coprime framework, applying a sliding translation technique to optimize sensor placement. Crucially, this rearranged structure preserves the continuity of the difference co-array (DCA). Furthermore, the sum co-array (SCA) is shifted to merge seamlessly with the DCA, eliminating redundancy and substantially expanding both the virtual aperture array (VAA) and the DOF. Consequently, the proposed array demonstrates superior performance in practical DOA estimation tasks involving non-circular signals. Simulation results and comparative analyses confirm that, relative to traditional Nested Arrays (NA), Extended Sliding Nested Array (ESNA), and other benchmark structures, the proposed array achieves better DOF and VAA, leading to enhanced estimation accuracy in practical scenarios.

Index Terms—Direction of arrival estimation, Non-circular signal, Nested array, Sum difference co-array, Degrees of freedom

I. INTRODUCTION

With the increasingly widespread application of direction of arrival (DOA) in military and commercial fields, it has also attracted more and more attention in array signal processing [1]–[7]. Although the traditional subspace algorithms based on uniform linear arrays can solve the problem, they have significant limitations. Firstly, underdetermined DOA estimation cannot be achieved. That is, for an array composed of N sensors, at most $N - 1$ source signals can be estimated. Secondly, the separation of the array sensors is constrained to

half a wavelength, which impacts the aperture enlargement of the array and introduces the consequences of mutual coupling effects.

To solve the above problems, sparse arrays have gradually come into the view of researchers, and their research has developed rapidly [8]–[13]. Minimum redundancy arrays (MRA), coprime array (CPA) and nested arrays (NA) are examples of traditional sparse arrays. The MRA array can break through the sensor spacing of half wavelength and achieve underdetermined DOA estimation. However, no closed-form expression can be given for its sensor position. The NA array not only retains its advantages but also can obtain the closed-form expression of its sensor position, thus receiving much attention in the research of sparse arrays. In addition to the classic NA arrays, a series of variants of NA arrays such as augmented nested arrays (ANAs) [13], coprime nested array (CNA) [14], super nested arrays (SNAs) [15], and generalized nested array (GNA) [16] have emerged, further improving the accuracy of DOA estimation.

Most of the arrays mentioned above are studies for circular signals. In recent years, research on DOA estimation for non-circular signals has also developed rapidly [17]–[19]. Two relocating sparse nested array (RSNA-I) structures, RSNA-I and RSNA-II, were presented in [17]. They can, respectively, reach the degrees of freedom (DOF) of $4M_1M_2 + 2(M_1 + M_2) + 1$ and $4M_1M_2 + 2M_1(2 - \text{mod}(M_1, 2)) + 2M_2 + 1 - \text{mod}(M_1, 2)$. In [18], the extended sliding nested array (ESNA) array was designed based on the sliding strategy and can achieve $2N_1N_2 + 2(N_1 + N_2) - 2J + 1 + 4 \lfloor (N_1N_2 + N_2 + 2J)/2 \rfloor$

DOF, where $J = \lceil N_1/2 \rceil - 1$.

This paper begins by laying out the fundamental principles of the sum-difference co-array (SDCA) concept. Building on this foundation, we've developed and put forward a novel nested array employing a sliding strategy, specifically tailored to handle DOA estimation for non-circular signals. Through the analysis and numerical simulation, and compared with NA, ESNA, and RSNA-I arrays. The findings indicate that the array developed in this research has superior accuracy of DOA estimation.

The following are the paper's primary contributions:

- 1) We design a sliding extended coprime nested array utilizing a sliding methodology. Provide the closed-form expression for the sensor position of the array and the generic formula for DOF.
- 2) Through the analysis and numerical simulation, compared with NA, ESNA, and RSNA-I arrays, the novel array configuration detailed in this research paper boasts a better DOF, which translates to more accuracy of DOA estimation outcomes while demonstrating superior performance in challenging low signal-to-noise ratio (SNR) scenarios.

The remaining structure of the paper is as follows: The section II introduces the array receiving model of non-circular signals and the concept of SDCA. The section III presents the array structure and specific details designed in this paper. Section IV is numerical simulation and analysis. The conclusion is given in Section V.

Notation: The bold black symbol $\mathbf{B}$ is represented as a matrix. Correspondingly, $\mathbf{A}^T$, $\mathbf{A}^*$, and $\mathbf{A}^H$ represent the transpose operation, conjugation, and conjugate transpose of the matrix, respectively. $\mathbf{A}_{i,j}$ represents the element located at row i and column j in the matrix. $E[\cdot]$ represents the expectation operation. $\text{diag}\{\mathbf{A}\}$ represents the vectorization of the diagonal elements of $\mathbf{A}$. $\odot$ denotes the Khatri-Rao product. $\langle i, j \rangle$ represents the generation of a uniform set of numbers from i to j . $\lceil a \rceil$ and $\lfloor a \rfloor$ represent the ceiling and floor integer values of a , respectively.

II. THEORETICAL PREMISE

A. Receiving signal model

Suppose there are Q signals incident on a receiving array in space. The signal satisfies the assumption of a far-field narrowband signal, with the incident direction being $\{\theta_1, \theta_2, \dots, \theta_Q\}$. The receiving array comprises M sensors, with their positions denoted by $\mathbb{W}$.

$$\mathbb{W} = \{w_1d, w_2d, \dots, w_Md\} \quad (1)$$

where d represents the minimum sensor spacing unit, with a size equal to half the wavelength of the incident signal. The sampling of the source signal by the array at time t can be modeled as:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{e}(t), \quad (2)$$

where $\mathbf{A} = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_Q)]$ is the matrix of the array manifold. $\mathbf{a}(\theta_q) =$

$[e^{j\pi w_1 d \sin(\theta_q)}, e^{j\pi w_2 d \sin(\theta_q)}, \dots, e^{j\pi w_M d \sin(\theta_q)}]^T$ is the direction vector corresponding to the source signal with an incident angle of θ_q . $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_Q(t)]^T$ and $\mathbf{e}(t) = [e_1(t), e_2(t), \dots, e_M(t)]^T$ represent the source signal and additive white noise respectively.

In this paper, we define the received signal as a non-circular signal and rerepresent it as [20] :

$$\mathbf{s}_{NC}(t) = \mathbf{\Psi}\mathbf{s}_R(t), \quad (3)$$

where $\mathbf{\Psi} = \text{diag}\{e^{-j\varphi_1}, e^{-j\varphi_2}, \dots, e^{-j\varphi_Q}\}$ represents the diagonal matrix of signal delay. φ_q represents the phase of the signal with an incident Angle of θ_q . When the number of snapshots is t , the signal vector is represented as $\mathbf{s}_R(t)$.

Note that the pseudo covariance matrix of the non-circular signals is nonzero [20]. In order to fully exploit the information conveyed by the non-circular signals, the combination of the original received signal and its conjugate is known as an extended received signal, and it can be written as follows:

$$\mathbf{x}_{NC}(t) = \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}^*(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A}\mathbf{\Psi} \\ \mathbf{A}^*\mathbf{\Psi}^* \end{bmatrix} \mathbf{s}_R(t) + \begin{bmatrix} \mathbf{e}(t) \\ \mathbf{e}^*(t) \end{bmatrix}. \quad (4)$$

The received signal's extended covariance matrix can be written as:

$$\begin{aligned} \mathbf{R}_{NC} &= E[\mathbf{X}_{NC}\mathbf{X}_{NC}^H] \\ &= \mathbf{A}_{NC}E[\mathbf{S}_R\mathbf{S}_R^H]\mathbf{A}_{NC}^H + \delta_n^2\mathbf{I}_{2M}, \\ &= \mathbf{A}_{NC}\mathbf{R}_{S_R}\mathbf{A}_{NC}^H + \delta_n^2\mathbf{I}_{2M} \end{aligned} \quad (5)$$

where $\mathbf{X}_{NC} = [\mathbf{x}_{NC}(1), \mathbf{x}_{NC}(2), \dots, \mathbf{x}_{NC}(T)]$. $\mathbf{A}_{NC} = [(\mathbf{A}\mathbf{\Psi})^T, (\mathbf{A}\mathbf{\Psi})^H]^T$ is defined as the array manifold matrix of the extended signal of dimension $2M \times Q$. $\mathbf{R}_{S_R} = E[\mathbf{S}_R\mathbf{S}_R^H] = \text{diag}\{\delta_1^2, \delta_2^2, \dots, \delta_Q^2\}$ is the covariance matrix of the source signal, δ_q^2 is the energy of the source signal in the q -th direction.

Vectorization of $\mathbf{R}_{NC}$ yields the following equation as:

$$\begin{aligned} \mathbf{z} &= \text{vec}(\mathbf{R}_{NC}) = (\mathbf{A}_{NC}^* \odot \mathbf{A}_{NC})\mathbf{s} + \delta_n^2 \text{vec}(\mathbf{I}_{2M}) \\ &= \mathbf{B}_{NC}\mathbf{s} + \delta_n^2 \text{vec}(\mathbf{I}_{2M}) \end{aligned}, \quad (6)$$

where $\mathbf{s} = [\delta_1^2, \delta_2^2, \dots, \delta_Q^2]^T$. $\mathbf{B}_{NC}$ is the array manifold matrix of the SDCA.

However, the phase information in $\mathbf{B}_{NC}$ is disordered, so a selection matrix $\mathbf{P}$ is constructed to organize the phase information in $\mathbf{B}_{NC}$.

$$\mathbf{z}_U = \mathbf{P}\mathbf{z} = \mathbf{B}'_{NC}\mathbf{s} + \delta_n^2 \mathbf{P}\text{vec}(\mathbf{I}_{2M}). \quad (7)$$

At this point, $\mathbf{z}_U$ can be thought of as the received signal of an array that has L_U sensors. L_U is the number of array elements of the maximum continuous segment of the SDCA virtual array of the original array. However, its snapshot number is 1. The full-rank covariance matrix is obtained by applying the spatial smoothing procedure for $\mathbf{z}_U$ based on [8]. The off-grid sparse Bayesian inference (OGSBI) approach is then used to estimate DOA [21].

B. Sum difference co-array theoretical models

To enable an array with a specific quantity of sensors to achieve better DOF for underdetermined DOA estimation, extending the length of its virtual array with Sum co-array (SCA) and Difference co-array (DCA) is an effective approach.

Definition 1: For an array whose sensor locations set is $\mathbb{W}$, the sensors locations set $\mathbb{W}_{SCA}$ of its SCA is defined as follows:

$$\mathbb{W}_{SCA} = \{\pm(\mathbb{W}_i + \mathbb{W}_j) \mid \mathbb{W}_i \in \mathbb{W}, \mathbb{W}_j \in \mathbb{W}\}. \quad (8)$$

Definition 2 : For an array with sensor locations denoted as $\mathbb{W}$, the sensor locations set $\mathbb{W}_{DCA}$ of its DCA is defined as follows:

$$\mathbb{W}_{DCA} = \{\mathbb{W}_i - \mathbb{W}_j \mid \mathbb{W}_i \in \mathbb{W}, \mathbb{W}_j \in \mathbb{W}\}. \quad (9)$$

Definition 3 : For an array whose sensor locations set is $\mathbb{W}$, the sensors locations set $\mathbb{W}_{SDCA}$ of its SDCA is defined as follows:

$$\mathbb{W}_{SDCA} = \mathbb{W}_{SCA} \cup \mathbb{W}_{DCA}. \quad (10)$$

In subsequent research, DOF specifically refers to the length of the maximum continuous segment of the SDCA in an array. Virtual aperture array (VAA) specifically refers to the SDCA length of the array, that is, $VAA = \max(\mathbb{W}_{SDCA}) - \min(\mathbb{W}_{SDCA})$.

III. THE PROPOSED SLIDING EXTEND COPRIME NESTED ARRAY

A. Structure of the Sliding Extend Coprime Nested Array

A sliding extended nested array (SECNA) based on coprime sets will be presented in this section. Firstly, the physical position of the array sensor is given, and then the closed-form expression of the DOF of its SDCA is presented.

The SECNA array consists of three subarrays, and the specific position expressions are as follows:

$$\mathbb{S} = \bigcup_{p=1,2,3} \text{Array}(p), \quad (11)$$

$$\begin{cases} \text{Array}(1) = N \langle 0, M-1 \rangle + S \\ \text{Array}(2) = M \langle 0, N \rangle + S \\ \text{Array}(3) = (N+M) \langle 1, N+M-1 \rangle + MN + S \end{cases}, \quad (12)$$

where N and M are a set of coprime numbers. $S = (M+N)^2/2$ is the horizontal sliding length using the sliding strategy. There are $2(M+N) - 1$ physical sensors in total.

The SECNA structure when the coprime set is $M = 5$ and $N = 3$ is depicted in Fig. 1. The three subarrays are respectively: $\text{Array}(1) = \{32, 35, 38, 41, 44\}$, $\text{Array}(2) = \{32, 37, 42, 47\}$ and $\text{Array}(3) = \{55, 63, 71, 79, 87, 95, 103\}$. There are fifteen physical sensors in all. As can be seen from Fig. 1(b), the length of the maximum continuous segment of the positive half-axis of the virtual SDCA array of this array is 142. Therefore, the DOF of the virtual SDCA array composed of 15 physical sensors in the SECNA array can reach 285.

Definition 4: The degree of freedom (DOF) of the SECNA array is the maximum continuous segment length of its SDCA array, specifically:

$$DOF = 4(M+N)^2 + 2MN - 1, \quad (13)$$

B. Comparison of degrees of freedom

This subsection presents a comparison between the DOF of the NA, ESNA, and RSNA-I arrays and the SECNA array suggested in this paper.

TABLE I
DOF COMPARISON OF DIFFERENT ARRAYS

Numbers of sensors	NA	ESNA	RSNA-I	SECNA
9	61	57	99	111
13	113	121	195	219
19	221	427	399	441
23	313	609	575	645
27	421	823	783	873

TABLE II
COUPLING LEAKAGE COEFFICIENT COMPARISON OF DIFFERENT ARRAYS

Numbers of sensors	NA	ESNA	RSNA-I	SECNA
13	0.3203	0.2697	0.2689	0.2336
15	0.3236	0.2804	0.2802	0.2072
19	0.3286	0.2942	0.2956	0.1923
25	0.3337	0.3074	0.3097	0.1591

Table I presents the specific values of DOF for different arrays, with the number of sensors ranging from 9 to 27. Table I makes it fairly evident that the DOF of the SDCA of various arrays varies depending on the number of sensors. Among them, the DOF of ESNA and RSNA-I arrays each have their advantages and disadvantages when the number of elements of array is different. However, the DOF of the SECNA array designed and proposed in this paper are superior to those of NA, ESNA and RSNA-I when the number of sensors varies. Furthermore, Table II presents a comparison of the coupling leakage coefficients for different arrays. SECNA exhibits the lowest coupling leakage coefficient regardless of the number of sensors, indicating better robustness to mutual coupling effects. Therefore, the SECNA array is bound to have better working performance when actually conducting DOA estimation.

IV. NUMERICAL SIMULATION

In this section, the performance of DOA estimation of different array structures are compared by numerical simulation in terms of root mean square error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{QK} \sum_{k=1}^K \sum_{q=1}^Q (\hat{\theta}_{qk} - \theta_q)^2}, \quad (14)$$

where K is the total number of Monte Carlo simulation trials. $\hat{\theta}_{qk}$ is the DOA estimate of the signal from direction θ_q in

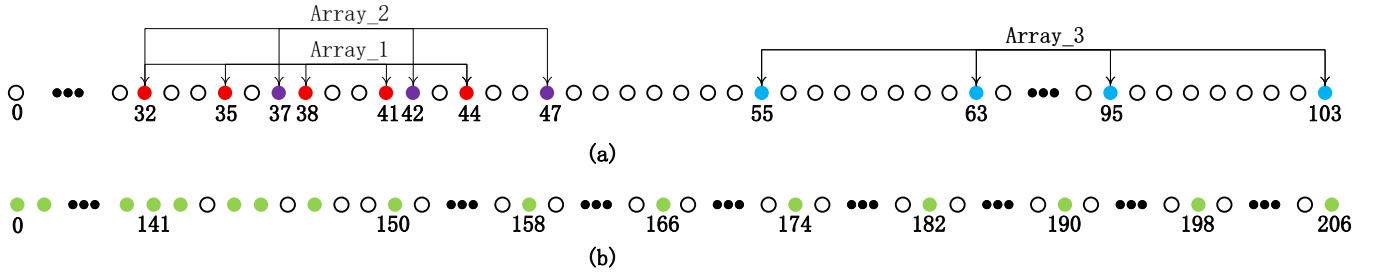


Fig. 1. (a) The SECNA array structure with $M = 5$ and $N = 3$; (b) Virtual SDCA on the positive side.

the k Monte Carlo simulation trials. In all simulations in this section, K is set to 200 runs.

The SECNA array created in this study will be compared with the current arrays (including NA, ESNA, and RSNA-I) in terms of DOA simulation performance. 13 physical sensors are included in the array. The estimated number of DOA sources is 31, with an angle ranging from -60° to 60° , following a uniform distribution. Furthermore, the OGSBI method is used in all of the simulations in this research.

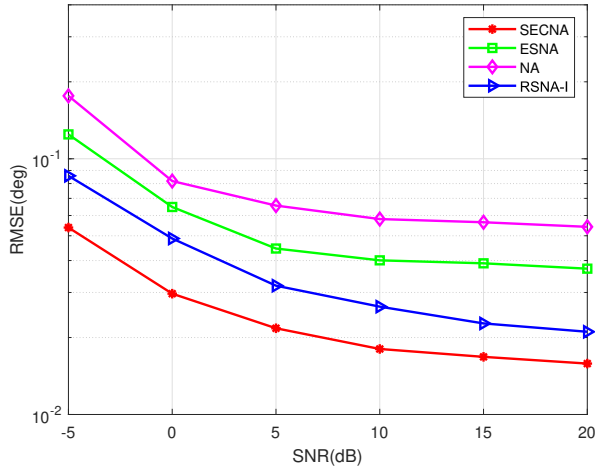


Fig. 2. RMSE versus SNR

The comparison of the RMSE of various arrays under various SNRs can be seen in Fig. 2. The snapshot count is established at 2000. The SNR range is established between -5dB and 20dB. It is evident that the RMSE of the SECNA array is much lower than that of other arrays under various SNR. RMSE continues to decline as SNR rises. This shows that SECNA has more prominent DOA estimation performance and stronger anti-noise ability.

The RMSE of various arrays under various snapshots is compared in Fig. 3. The SNR is set to 20dB. The snapshots range is set to 300 to 2600. It is evident that the SECNA array's RMSE is substantially lower than those of other arrays under various snapshots. As snapshot increases, RMSE further decreases. This shows that SECNA has more prominent DOA estimation performance.

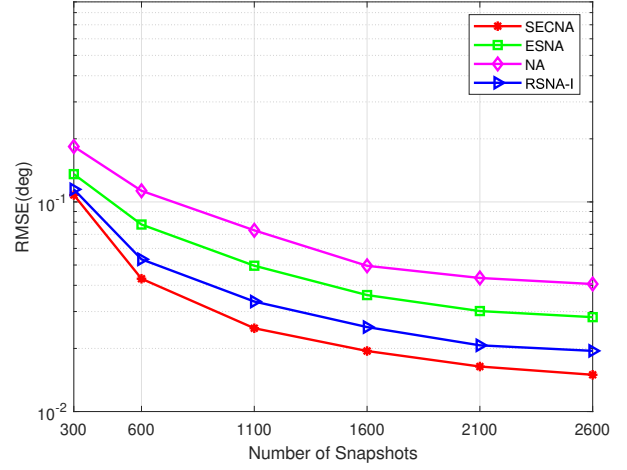


Fig. 3. RMSE versus snapshots

V. CONCLUSION

In this paper, we proposed a novel nested array. better DOF are achieved by using the SDCA in this array, which is based on the sliding strategy. Through numerical analysis and simulation verification, the proposed array has a better DOF and superior estimate performance in DOA estimation where the number of physical sensors is equal to that of NA, ESNA, and RSNA-I arrays.

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